

PAYGO payroll tax: two-rate diagnostic

Same selected parameters and utility scale; endogenous balanced pension and housing equilibrium

Outcome	17.9% rate	8.751% rate	Change
Payroll rate	17.900%	8.751%	-9.149 pp
Pension / mean annual gross earnings	51.159%	25.011%	-26.148 pp
Payroll revenue (four-year units)	0.530	0.259	-0.271
Pension outlays (four-year units)	0.530	0.259	-0.271
Mean working disposable income	0.821	0.912	+0.091
Aggregate wealth	3.883	5.380	+1.498
Ownership, ages 30-55	39.493%	41.881%	+2.387 pp
Mean occupied rooms	6.477	6.494	+0.018
House price	0.624	0.628	+0.005
Childlessness, ages 40-44	16.909%	12.538%	-4.371 pp
Mean age at first birth	26.658	25.942	-0.715
Top-coded children-ever-born stock	2.100	2.252	+0.152
Raw stationary children-state mean	1.883	2.012	+0.129

Finding: the proposal halves the pension and raises saving, ownership and fertility at fixed preferences.

It cannot select a tax rate: the target contract is frozen, and demographic entry is held fixed.

The 2.1 top-coded stock moves to 2.252. It equals adjusted birth flow / fixed entry flow here.

This is not a new demographic steady state, period total fertility rate, or cohort-completion measure.

Complete frozen target comparison

Selected September 23 target/weight contract, all 13 rows; contributions are descriptive

Moment	Target	Current	Proposal	Proposal gap	Weight	Contribution
initial normalization	2.100	2.100	2.252	0.152	-	-
cps childlessness	0.198	0.169	0.125	-0.073	35,532.304	188.821
cps exactly one	0.214	0.246	0.224	0.010	26,952.821	2.831
nchs mean age	25.976	26.658	25.942	-0.034	139.828	0.160
nchs share30	0.249	0.266	0.221	-0.028	13,866.065	10.903
wealth earnings	6.146	5.241	7.263	1.117	7.595	9.478
bequest wealth	0.009	0.007	0.007	-0.002	5,165,289.256	25.310
old dispersion	3.516	4.099	2.983	-0.533	10.616	3.016
mean rooms	5.561	6.477	6.494	0.933	128.021	111.478
ownership 30 55	0.648	0.395	0.419	-0.230	2,339.362	123.244
first birth rooms	0.720	0.987	1.023	0.302	137.565	12.586
family rooms	0.347	0.283	0.288	-0.059	280.528	0.993
recent parent ownership	0.163	0.125	0.122	-0.040	27,055.823	44.312

The separate normalization row has no weight. Weighted contributions do not choose the policy rate.

Full-precision values, current gaps and current contributions are in full_target_comparison.csv.

Complete parameter and restriction table

Nine selected structural coordinates are held fixed; payroll rate changes and pension balances endogenously

Parameter	Current	Proposal	Lower	Upper	Near bound
beta_annual	0.976	0.976	0.940	0.990	no
kappa_fert	0.209	0.209	0.020	50.000	yes
kappa_fert_continuation	0.482	0.482	0.020	50.000	yes
chi	0.888	0.888	0.100	5.000	no
H0	8.967	8.967	0.200	80.000	no
theta0	0.088	0.088	0.000	8.000	no
theta1	0.088	0.088	0.020	16.000	yes
first_birth_fixed_cost	0.353	0.353	0.000	8.000	no
h_P	2.044	2.044	0.100	2.300	no
hbar_child_rooms	0.000	0.000	-	-	-
psi_child	0.142	0.142	-	-	-
payroll_tax	0.179	0.088	-	-	-
pension_period	2.046	1.000	-	-	-
housing_supply_elasticity	0.630	0.630	-	-	-
tenure_choice_kappa	0.005	0.005	-	-	-
alpha_cons	0.733	0.733	-	-	-
sigma	2.000	2.000	-	-	-

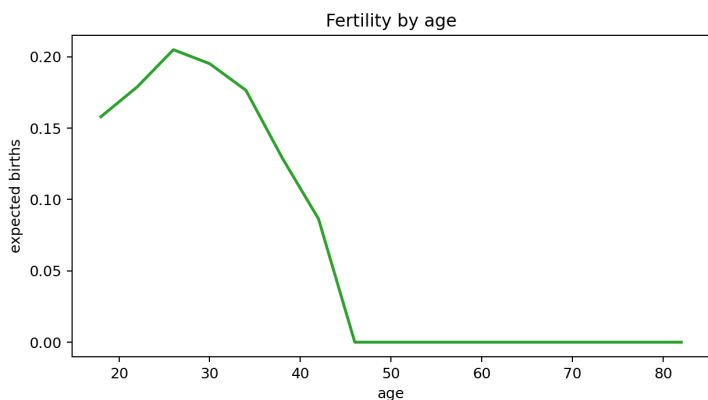
Bounds are the selected run's actual search restrictions; blank means externally fixed or derived.
The housing supply elasticity is 0.630. The September 24 maturation/entry decision is not implemented.

Standard diagnostic figures

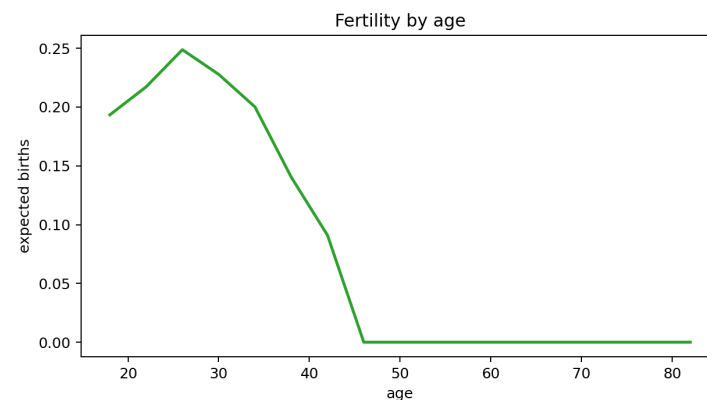
Matched current and proposed cases | figure pairs 1-2 of 17

fertility by age

Current: 17.9%

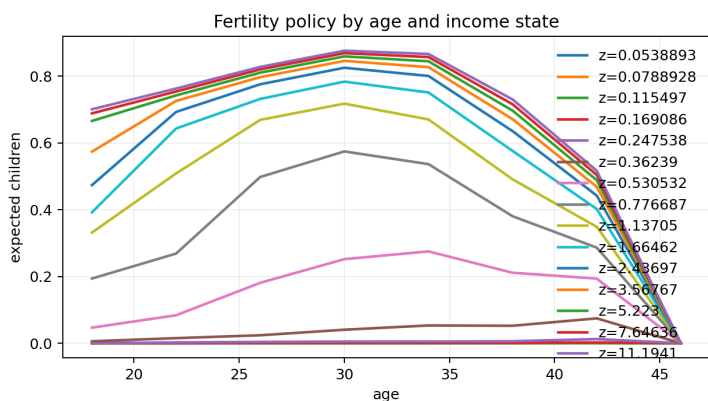


Proposal: 8.751%

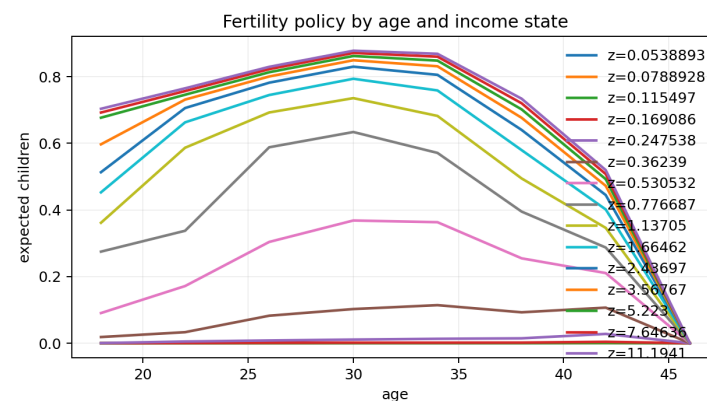


fertility policy by age income state

Current: 17.9%



Proposal: 8.751%



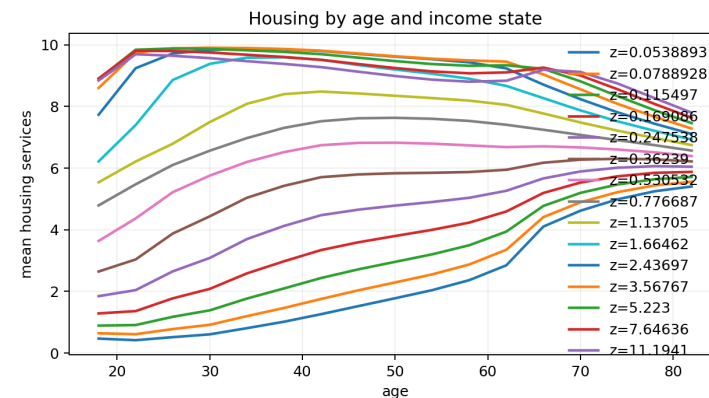
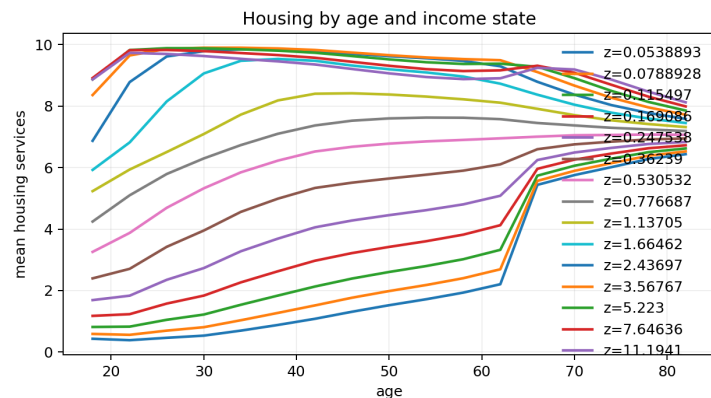
Standard diagnostic figures

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housing by age income state

Current: 17.9%

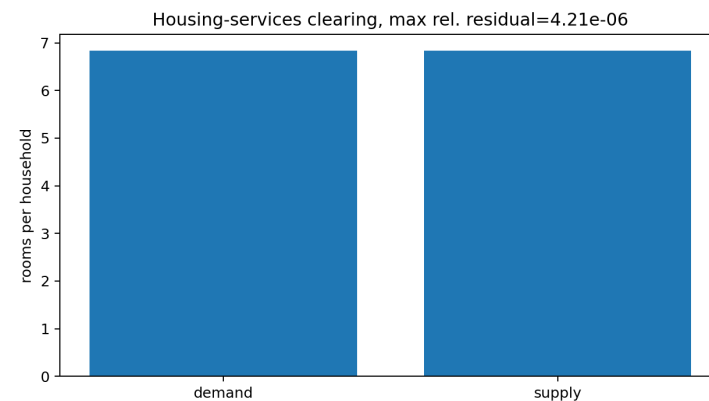
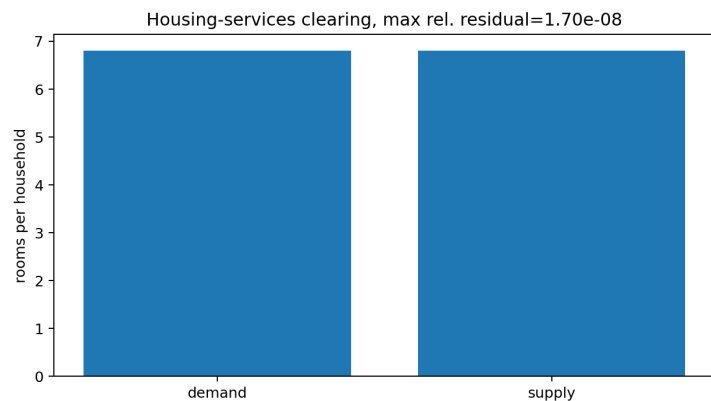
Proposal: 8.751%



housing market

Current: 17.9%

Proposal: 8.751%



Standard diagnostic figures

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housing prices

Current: 17.9%

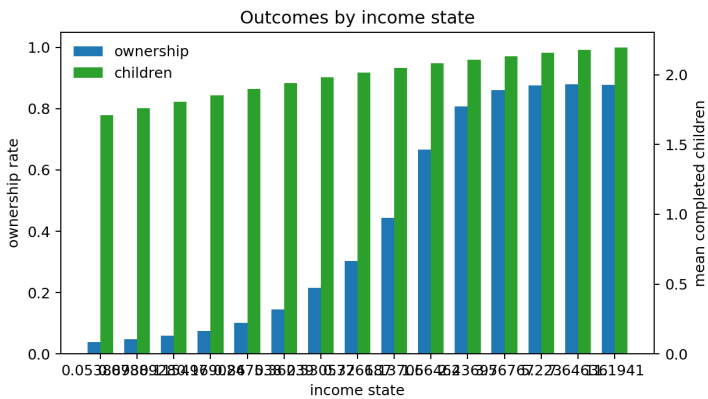
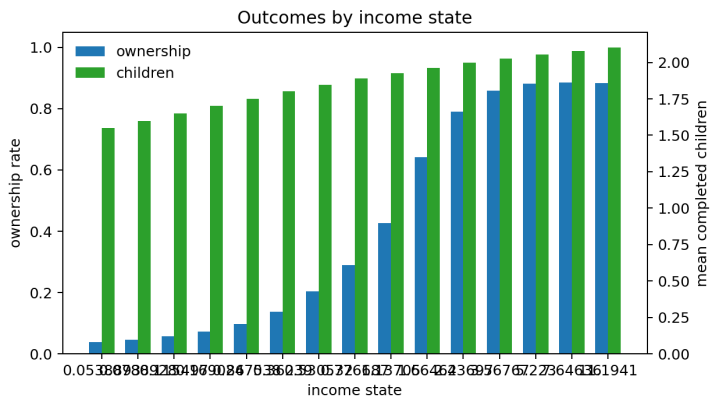
Proposal: 8.751%



income state outcomes

Current: 17.9%

Proposal: 8.751%



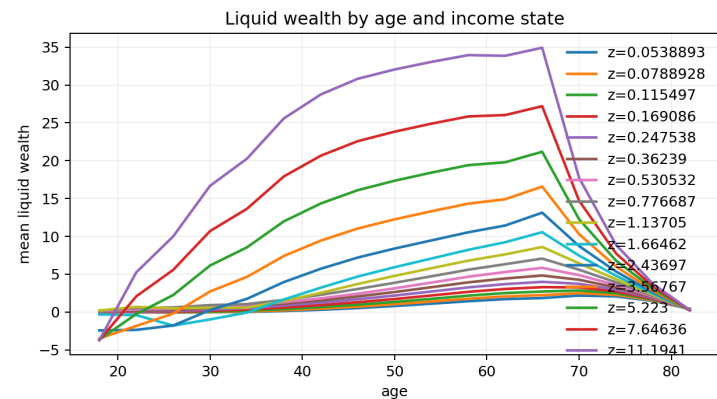
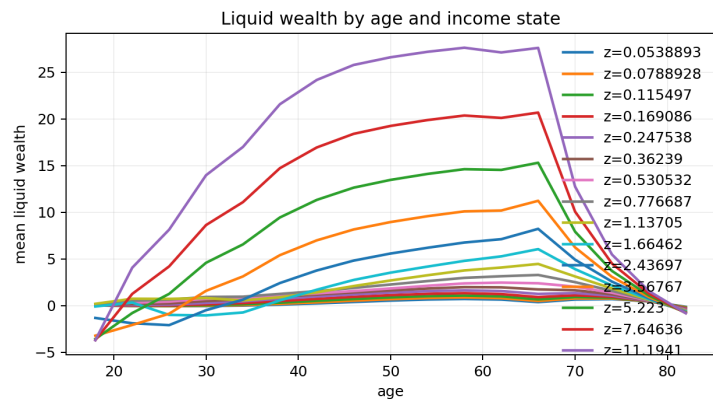
Standard diagnostic figures

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liquid wealth by age income state

Current: 17.9%

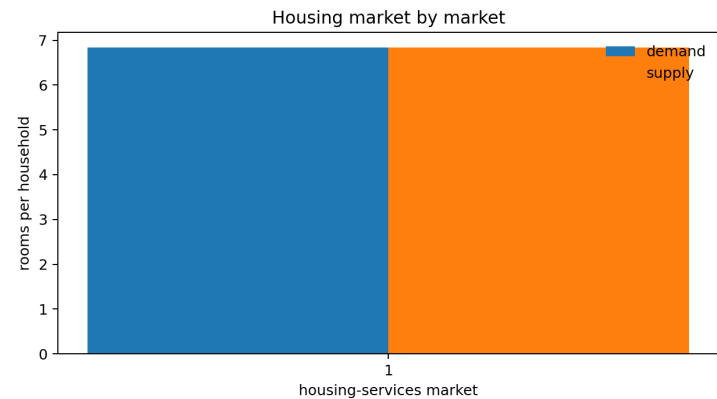
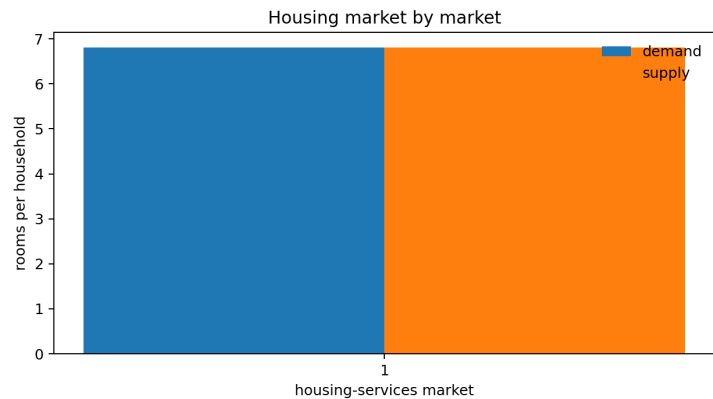
Proposal: 8.751%



market clearing by market

Current: 17.9%

Proposal: 8.751%



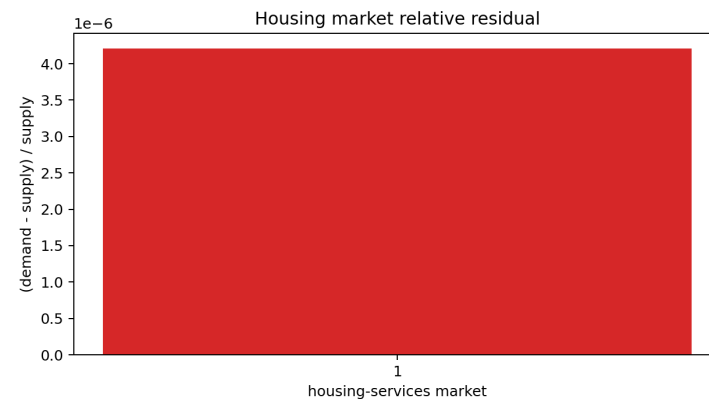
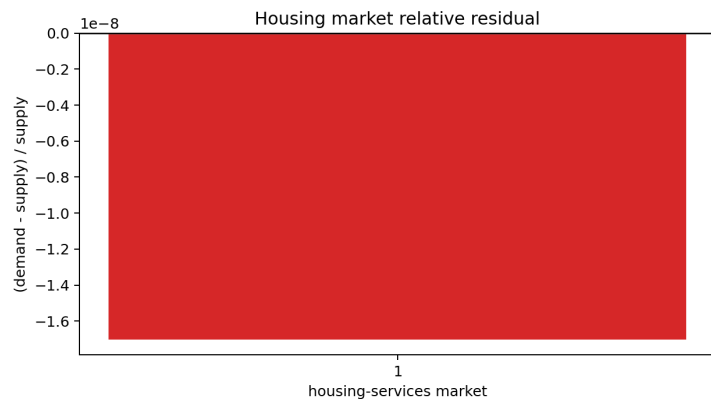
Standard diagnostic figures

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market clearing residuals

Current: 17.9%

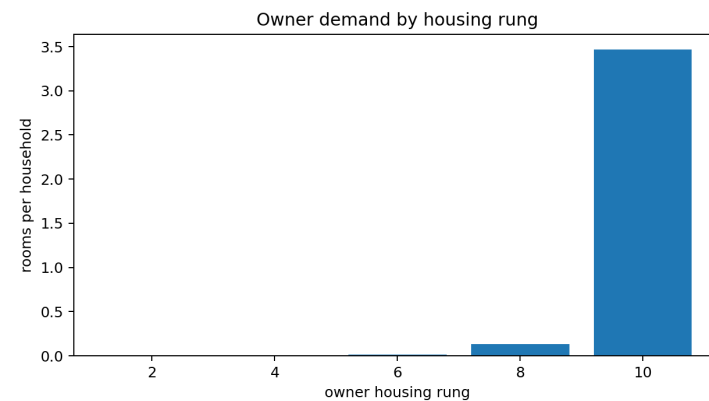
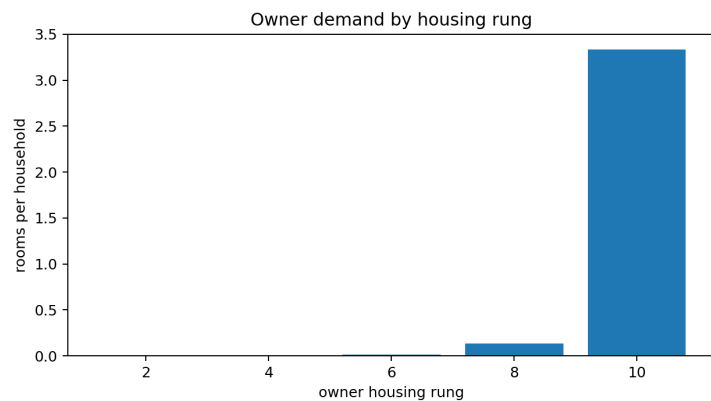
Proposal: 8.751%



owner rungs

Current: 17.9%

Proposal: 8.751%

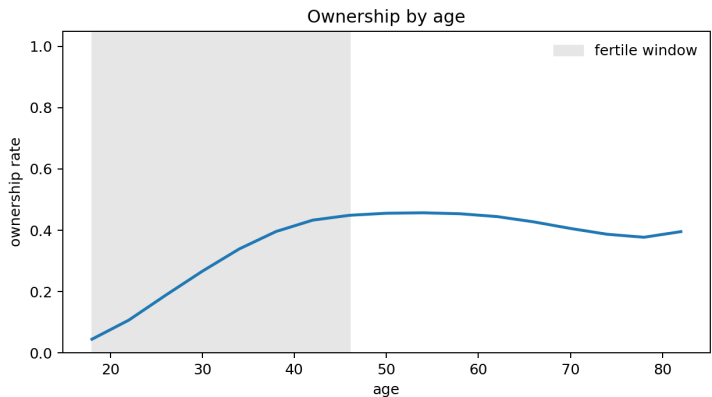


Standard diagnostic figures

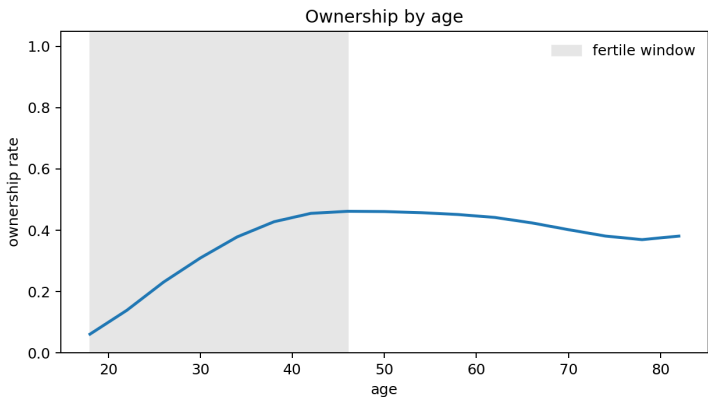
Matched current and proposed cases | figure pairs 11-12 of 17

ownership by age

Current: 17.9%

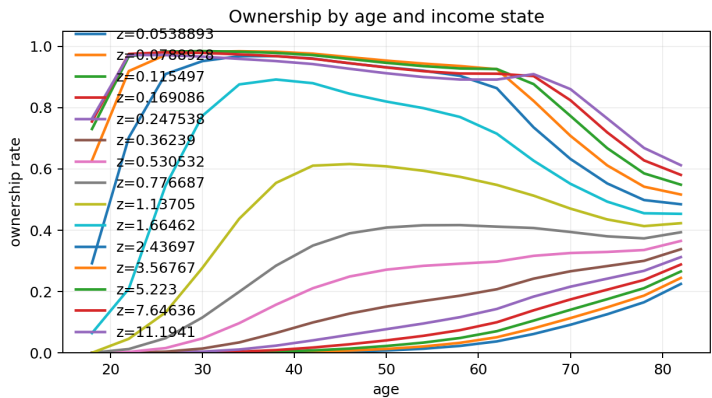


Proposal: 8.751%

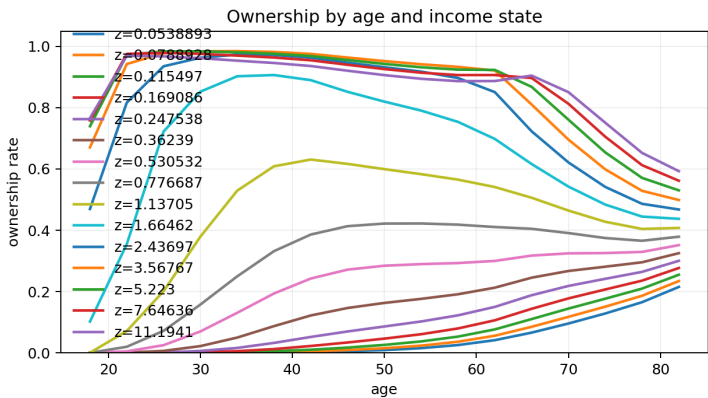


ownership by age income state

Current: 17.9%



Proposal: 8.751%



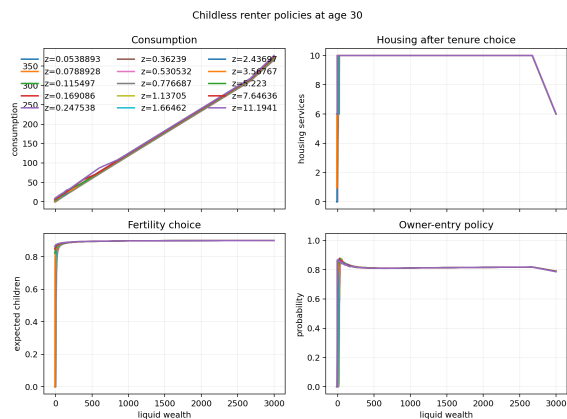
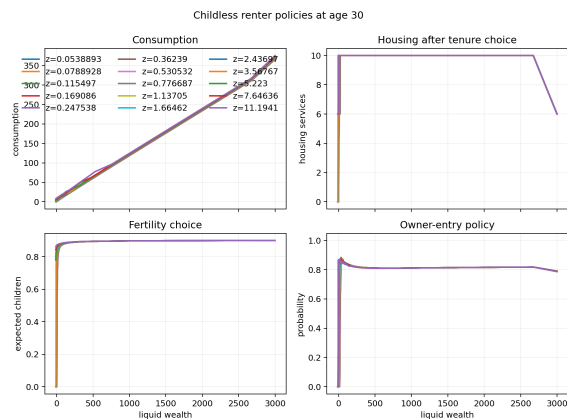
Standard diagnostic figures

Matched current and proposed cases | figure pairs 13-14 of 17

policy childless renter age30

Current: 17.9%

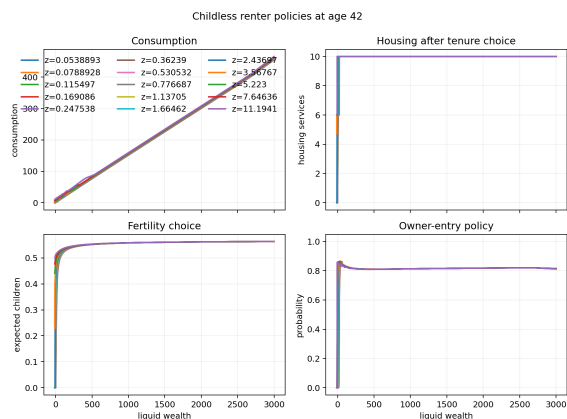
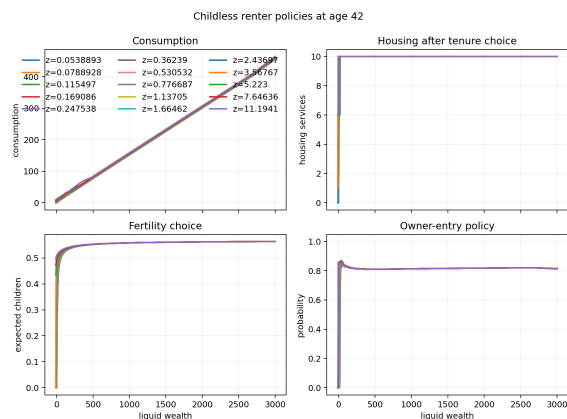
Proposal: 8.751%



policy childless renter age42

Current: 17.9%

Proposal: 8.751%



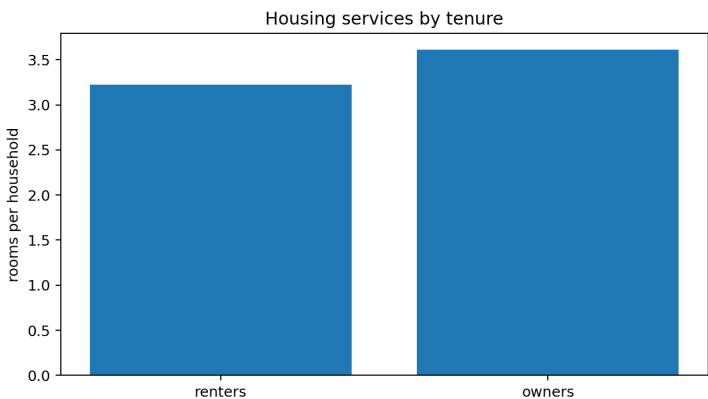
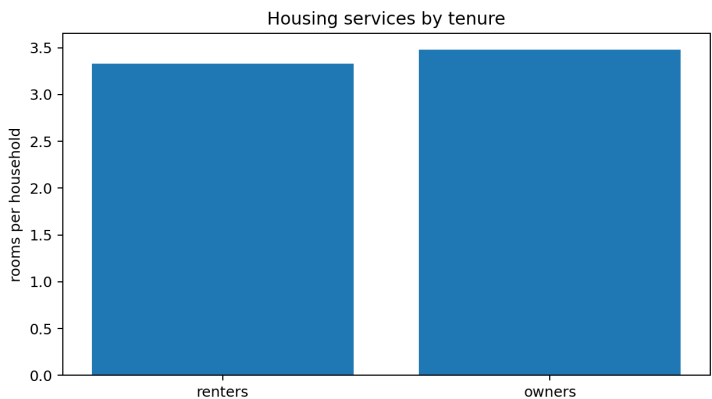
Standard diagnostic figures

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tenure services

Current: 17.9%

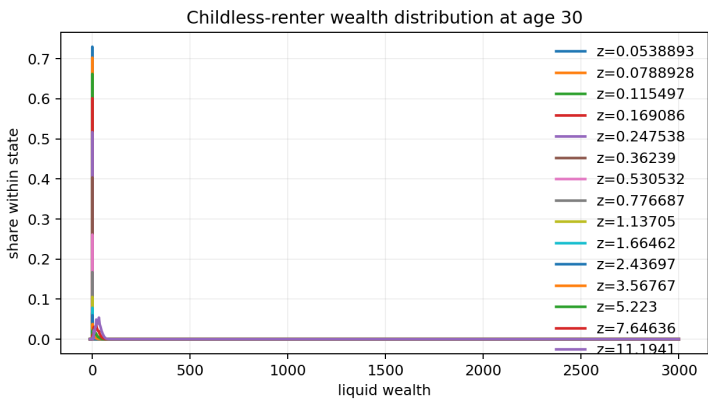
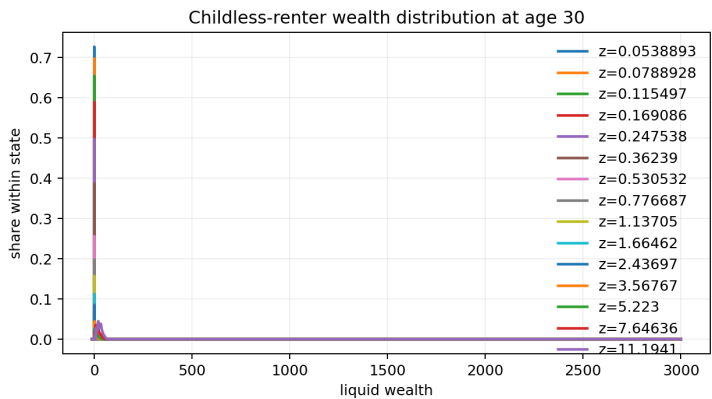
Proposal: 8.751%



wealth dist childless renter age30

Current: 17.9%

Proposal: 8.751%



Standard diagnostic figures

Matched current and proposed cases | figure pairs 17-17 of 17

wealth dist childless renter age42

Current: 17.9%

Proposal: 8.751%

